

ELAT, Israel

WMO: 401990

Lat: 29.5526N

Lon: 34.9520E

Elev: 72

StdP: 14.66

Time Zone: 2.00 (ISR)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.6	49.8	18.4	13.9	69.9	22.8	17.2	71.2	22.1	65.8	20.7	65.7	5.7	30	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	109.1	69.9	107.0	69.4	105.1	69.2	75.6	96.3	74.3	95.6	73.3	95.1	13.1	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	106.1	88.1	67.2	100.3	87.4	65.9	95.9	87.0	39.1	96.7	37.9	95.4	36.9	95.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	21.0	19.5	DB	42.6	113.6	2.2	2.2	41.1	115.2	39.8	116.5	38.5	117.7	37.0	119.3	(4)
(5)			WB	36.9	79.0	2.6	1.5	35.0	80.0	33.5	80.9	32.1	81.7	30.2	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	78.6	61.0	64.2	70.0	77.0	85.0	90.4	93.5	93.4	89.0	82.0	72.4	63.8	(6)
(7)		DBStd	12.37	4.00	5.09	5.65	5.91	5.25	3.87	2.78	2.68	3.19	4.66	5.03	4.30	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	299	135	69	14	1	0	0	0	0	0	0	5	75	(9)
(10)		CDD50	10426	340	397	621	811	1086	1213	1349	1345	1171	993	673	427	(10)
(11)		CDD65	5252	11	46	171	362	621	763	884	880	721	528	228	37	(11)
(12)		CDH74	74808	78	356	1501	3986	8613	12086	14692	14545	10774	6216	1744	217	(12)
(13)		CDH80	46156	5	76	494	1858	5078	8000	10249	10097	6661	3079	526	33	(13)
(14)	Wind	WSAvg	9.4	7.5	8.1	9.0	9.4	9.8	10.8	9.7	10.6	11.1	9.8	8.8	7.9	(14)
(15)	Precipitation	PrecAvg	0.9	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2	(15)
(16)		PrecMax	2.4	0.9	2.6	0.9	1.1	0.5	0.0	0.0	0.0	0.0	1.4	1.1	1.7	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.6	0.2	0.4	0.2	0.2	0.1	0.0	0.0	0.0	0.0	0.3	0.2	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.2	86.4	93.6	100.7	107.9	111.3	111.4	110.9	107.2	102.2	92.9	84.3	(19)
MCWB			57.7	60.0	62.0	63.3	67.4	68.8	70.8	72.5	70.7	67.9	64.9	60.1	(20)	
2%		DB	75.8	82.2	89.4	96.7	103.9	107.8	109.0	107.8	103.9	98.2	88.6	78.9	(21)	
		MCWB	56.8	58.1	60.7	63.0	66.8	68.2	70.3	70.7	70.8	68.3	63.6	58.8	(22)	
5%		DB	73.3	78.4	85.7	93.2	100.5	105.4	106.5	105.8	101.7	94.9	85.8	75.8	(23)	
		MCWB	56.1	57.4	59.9	62.0	65.6	67.8	69.5	70.1	70.4	68.1	62.9	57.9	(24)	
10%		DB	70.7	75.1	82.0	89.8	97.2	102.5	104.4	104.0	99.0	92.0	82.6	73.5	(25)	
		MCWB	54.8	56.6	58.9	61.4	64.8	67.6	69.3	70.4	70.1	67.6	61.9	56.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	66.5	67.9	69.0	71.7	73.6	76.8	77.9	77.0	74.4	69.3	67.0	(27)
MCD B			73.2	77.1	79.7	86.0	92.0	96.2	97.6	99.2	96.6	90.4	83.5	76.7	(28)	
2%		WB	59.4	61.9	64.7	66.5	69.6	71.9	74.7	75.8	75.0	72.7	66.8	61.5	(29)	
		MCD B	72.3	76.5	81.3	86.9	94.1	96.4	97.3	97.1	95.1	90.0	83.6	75.5	(30)	
5%		WB	57.0	59.2	62.3	64.5	68.1	70.6	73.3	74.5	73.6	71.1	65.0	59.2	(31)	
		MCD B	71.0	75.1	81.0	87.5	94.0	96.5	97.4	96.2	93.7	88.1	81.5	72.8	(32)	
10%		WB	55.4	57.2	60.2	62.9	66.8	69.5	72.1	73.4	72.4	69.7	63.5	57.4	(33)	
		MCD B	68.7	73.1	79.6	86.3	92.6	96.3	96.5	95.7	92.6	87.1	79.6	71.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.2	20.3	20.8	22.1	23.3	24.3	23.4	22.4	21.3	20.0	19.9	19.7	(35)
MCD BR			22.5	24.8	25.0	25.5	26.0	26.9	25.1	24.0	23.2	22.4	22.2	22.3	(36)	
MCWBR			11.1	11.1	10.8	9.7	9.0	7.9	7.6	7.2	7.6	8.4	9.8	10.8	(37)	
5% WB		MCD BR	20.0	21.5	21.8	22.1	23.2	24.1	22.7	21.4	21.1	19.7	19.8	19.6	(38)	
		MCWBR	11.2	11.5	11.5	10.1	9.5	8.3	8.4	7.8	8.0	8.4	9.5	11.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.405	0.467	0.494	0.499	0.420	0.425	0.433	0.430	0.436	0.373	0.346	(40)	
taud		2.251	2.108	1.913	1.876	1.892	2.152	2.168	2.153	2.148	2.110	2.272	2.340	(41)		
Ebn at Noon		272	270	259	257	256	275	274	271	267	256	266	273	(42)		
Edh at Noon		36	46	60	64	63	49	48	48	47	45	36	32	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1216	1519	1951	2270	2496	2703	2612	2425	2121	1700	1348	1147	(44)	
(45)		RadStd	50	58	63	51	81	28	42	33	48	51	40	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.65	N/A	-2.27	N/A	N/A	N/A	N/A	N/A	+243	+263
(47)	Regional (11 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+263

Nomenclature: See separate page